



TSegLab: Multi-stage 3D dental scan segmentation and labeling

Ahmed Rekik ^{a,c,b}, Achraf Ben-Hamadou ^{a,c}, Oussama Smaoui ^e, Firas Bouzguenda ^e, Sergi Pujades ^d, Edmond Boyer ^d

^a Digital Research Center of Sfax, Technopark of Sfax, Sakiet Ezzit, 3021 Sfax, Tunisia

^b ISSAT, Gafsa university, Sidi Ahmed Zarrouk University Campus, 2112 Gafsa, Tunisia

^c Laboratory of Signals, systems, artificial Intelligence and networks, Technopark of Sfax, Sakiet Ezzit, 3021 Sfax, Tunisia

^d Inria, Univ. Grenoble Alpes, CNRS, Grenoble INP, LJK, France

^e Udini, 37 BD Aristide Briand, 13100 Aix-En-Provence, France

ARTICLE INFO

Keywords:

Dental scan segmentation
Teeth classification
Teeth segmentation
Graph neural network
3D intraoral scan
Teeth3DS

ABSTRACT

This study introduces a novel deep learning approach for 3D teeth scan segmentation and labeling, designed to enhance accuracy in computer-aided design (CAD) systems. Our method is organized into three key stages: coarse localization, fine teeth segmentation, and labeling. In the teeth localization stage, we employ a Mask-RCNN model to detect teeth in a rendered three-channel 2D representation of the input scan. For fine teeth segmentation, each detected tooth mesh is isomorphically mapped to a 2D harmonic parameter space and segmented with a Mask-RCNN model for precise crown delineation. Finally, for labeling, we propose a graph neural network that captures both the 3D shape and spatial distribution of the teeth, along with a new data augmentation technique to simulate missing teeth and teeth position variation during training. The method is evaluated using three key metrics: Teeth Localization Accuracy (TLA), Teeth Segmentation Accuracy (TSA), and Teeth Identification Rate (TIR). We tested our approach on the Teeth3DS dataset, consisting of 1800 intraoral 3D scans, and achieved a TLA of **98.45%**, TSA of **98.17%**, and TIR of **97.61%**, outperforming existing state-of-the-art techniques. These results suggest that our approach significantly enhances the precision and reliability of automatic teeth segmentation and labeling in dental CAD applications. Link to the project page: <https://crns-smartvision.github.io/tseglab>

1. Introduction

In the recent years, the field of Computer-Aided Dentistry (CAD) has advanced rapidly, taking advantage of both hardware and software progress. Intraoral 3D scanners, for instance, are increasingly being used to enable accurate diagnosis and treatment planning. Dentists using CAD systems often rely on 3D dental scans to add, move, or reshape teeth in order to simulate orthodontic and/or restorative treatments. In this regard, 3D teeth segmentation and classification are essential for CAD systems. Before beginning any treatment, the CAD system must accurately segment and classify each tooth in the 3D dental scan to avoid time-consuming manual adjustments by the dentist and enable fully automated procedures. Automatic segmentation, which takes only a few seconds, can eliminate manual tasks that typically take around 30 min per scan, thereby accelerating clinical dental workflows.

Despite prior research efforts to enhance 3D tooth segmentation performance, the development of a robust and accurate method remains challenging due to various factors. First, intraoral 3D scans (IOSs) exhibit significant variability between patients. Furthermore, some scans

may contain complex abnormalities (e.g., missing or crowded teeth), leading to misaligned teeth and irregular dental boundaries that affect tooth segmentation and instance separation.

Many previous works have been proposed to deal with these issues. Geometry-based methods rely on tooth geometric features such as curvature thresholding [1], active contour computing [2], or harmonic field [3] in order to separate the teeth from the gingiva. These approaches, however, are extremely sensitive to irregular tooth 3D shapes and their complex curvatures, which affects the segmentation results.

Recent advancements in machine learning-based 3D shape analysis are also being applied to the analysis of 3D dental scans. Two main approaches can be identified. Some researchers propose converting 3D scans into 2D images in order to use existing CNN architectures for both teeth segmentation and classification [4]. While this approach may produce satisfactory results in the 2D space, accurately reconstructing the segmentation in 3D space can be challenging, particularly in the accurate identification of the teeth boundaries. To address this, Lian et al. [5] proposed an end-to-end deep learning-based method (MeshSegNet)

* Corresponding author at: Digital Research Center of Sfax, Technopark of Sfax, Sakiet Ezzit, 3021 Sfax, Tunisia.

E-mail address: achraf.benhamadou@crns.rnrt.tn (A. Ben-Hamadou).

<https://doi.org/10.1016/j.combiomed.2024.109535>

Received 22 February 2024; Received in revised form 17 August 2024; Accepted 3 December 2024

Available online 20 December 2024

0010-4825/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

working directly in the 3D space for teeth segmentation and labeling. Although this method offers faster results, it must overcome several challenges such as point cloud irregularity, resampling invariance, and scan abnormalities.

Additionally, due to the high resolution of intraoral 3D scans (a large number of vertices and triangles), 3D deep learning-based methods often require a downsampling process for teeth localization and segmentation as demonstrated in [6,7]. However, this downsampling step can lead to a decreased performance, particularly in the segmentation task, as the results need to be restored back to the full resolution. Furthermore, working with the full resolution requires a larger GPU resources for both training and inference phases. Teeth shape shows important similarities within certain classes, such as the first/second molar and central/lateral incisors. Therefore, the task of teeth classification should take into account not only teeth shape but also the relative position between teeth. It is worth noting that both the shape and the relative position between teeth can be ambiguous. These two aspects are not explicitly addressed in the current state-of-the-art methods for teeth classification.

In this paper, we propose an efficient multi-stage deep learning approach for the segmentation and labeling of 3D teeth scans. Our approach benefits from separating the three main tasks: localization, segmentation, and labeling. This division allows to address each challenge independently, based on the specific needs of the final clinical application, without requiring the processing of all three tasks simultaneously. The teeth localization component leverages the recent advances in 2D object detection using CNNs to achieve a coarse, yet robust, localization of visible teeth in the scans. This step yields a set of teeth candidates cropped with a certain margin from 3D scans. Subsequently, individual 3D segmentation is performed, utilizing harmonic mapping and curvature information computed from the 3D scan data. The goal of this segmentation step is to accurately separate the targeted tooth from the gum and neighboring teeth, taking into account the shape characteristics derived from the harmonic mapping and curvature analysis. Once teeth are localized and segmented, the teeth classification task is designed as a graph neural network (GNN), modeling the shape appearance of each individual tooth as well as the spatial arrangement of all the teeth in the jaw. Our approach is evaluated on the publicly released Teeth3DS benchmark [8] and compared quantitatively and qualitatively with state-of-the-art methods.

The remaining sections of this article are organized as follows. Section 2 provides a brief overview of the recent history of 3D teeth segmentation and labeling. Then, our proposed approach is detailed in Section 3. Afterward, Section 4 discusses the experimental validation and the obtained results. Finally, in Section 5, we summarize our findings and suggest potential directions for future research.

2. Related work

Medical image analysis covers a vast amount of studies across different modalities and medical fields, including Magnetic Resonance Imaging (MRI) analysis [9], skin lesion analysis [10–12], and nuclei image analysis [13]. In the dental field, various imaging modalities are employed for different purposes, such as panoramic X-rays [14–17], Cone-Beam Computed Tomography (CBCT) [18], intraoral photos [19], and intraoral 3D scans [20,21]. Each of these modalities serves specific diagnostic and treatment needs, requiring tailored methods and approaches for analysis. For a more comprehensive overview of these modalities, the reader is referred to Shujaat et al. [22] and Chen et al. [23].

Within the scope of this study, we specifically focus on intraoral 3D scans, with an emphasis on the detection, segmentation, and labeling of teeth within this modality. There are two main types of approaches when it comes to segmenting and labeling intraoral 3D scans: handcrafted feature-based methods and learning-based methods. Most of the relevant literature falls into one of these two categories.

2.1. Handcrafted features-based approaches

Previous methods predominantly relied on the utilization of manually designed geometric characteristics for segmenting intraoral scans. These methods can be classified into three main categories: methods based on surface curvature, methods based on contour lines, and methods based on harmonic fields. Surface curvature plays a crucial role in providing valuable information for characterizing the intraoral scan surface and accurately identifying tooth/gum boundaries. Zhao et al. [24] utilized this feature to introduce a semi-automatic technique for teeth segmentation, employing curvature thresholding. The method involves isolating the gums first, followed by the identification of boundary curves of each tooth to assist the segmentation task. Another technique was developed by Yuan et al. [25] that combined different elements for modeling separate teeth in dental scans. This involved using surface minimum curvature to extract certain areas and distinguish teeth from one another. An alternative approach, presented by Kronfeld et al. [26], utilizes active contour-line models to detect the boundaries separating teeth and gingiva. In the contour-line method, users manually select landmark points along the tooth boundaries on dental 3D scans. These selected landmarks are subsequently employed to compute contour lines using geodesic information, ultimately producing the desired tooth boundaries. This technique has been implemented in studies conducted by Sinthanayothin and Tharanont [27] and Yaqi and Zhongke [2]. The harmonic-field approach could also be employed for teeth segmentation, offering a more user-friendly alternative to previous methods by requiring minimal user interaction. In [3,28], a manual selection of a small number of surface points is required before starting the segmentation process. These methods prove to be more efficient in terms of reducing both the time and complexity involved in the segmentation process when compared to other techniques.

The aforementioned approaches face certain limitations in achieving a fully automated and robust segmentation of intraoral 3D scans. Surface curvature-based methods encounter challenges in determining the optimal threshold value, and they remain sensitive to noise. Choosing an incorrect threshold can result in over- or under-segmentation, leading to accuracy issues. Moreover, the manual threshold selection process makes it challenging to apply curvature-based methods in a fully automated segmentation approach. In contrast, approaches based on contour lines demand substantial user input and are time-consuming, making them less user-friendly. Lastly, harmonic field-based methods entail sophisticated and computationally intensive pre-processing.

2.2. Learning based approaches

Recently, there has been a shift in teeth segmentation techniques from relying on handcrafted features to leveraging learned features through the advancements of deep learning techniques. It has become evident that utilizing data-driven feature extraction, such as CNNs, yields superior performance compared to handcrafted features in various computer vision applications such as object detection [29]. This trend also extends to the domain of 3D teeth segmentation and labeling. Learning-based methods can be categorized into two primary approaches, based on the input data: 2D image segmentation or 3D mesh segmentation.

Numerous studies have utilized CNNs to analyze and extract relevant characteristics from Cone-Beam Computed Tomography (CBCT) images [18]. Cui et al. [30] introduced a supervised neural network architecture specifically designed for the automatic segmentation and identification of individual teeth from CBCT images. In their approach, a set of edge maps were initially generated from the CBCT slices using a CNN autoencoder. Subsequently, a Mask R-CNN [31] was applied to these edge maps to achieve tooth segmentation and labeling. Additionally, Rao et al. [32] proposed a symmetric fully convolutional residual neural network that generates a segmentation probability map

in CBCT images. Subsequently, teeth boundary smoothing was then accomplished through the application of the dense conditional random field technique, while segmentation enhancement was achieved using a deep bottleneck architecture. More recently, Do Nascimento Gerhardt et al. [33] utilized the Virtual Patient Creator tool, an AI-driven solution based on a deep learning algorithm, to detect and segment teeth and small edentulous areas on CBCT using two sequential networks built on the 3D U-Net architecture. The study demonstrated that this tool is both accurate and efficient for CBCT-based detection, segmentation, and labeling of teeth and missing teeth in cases of partial edentulism. Furthermore, Zhang et al. [4] employed an isomorphic mapping technique to generate 2D harmonic parameter space from intraoral 3D scans. These generated 2D images were then utilized as input for a U-Net network, enabling tooth segmentation from images.

As the deep learning field advances and the computation capabilities of hardware increase, researchers began to leverage deep learning techniques for the analysis of intraoral 3D scans. In their work, Sun et al. [34] utilized FeaStNet [35], a graph-based convolutional neural network, for tooth segmentation and recognition from intraoral 3D scans. Subsequently, an extension was made to the previous architecture by proposing an end-to-end graph convolutional network-based model for tooth segmentation and achieving dense correspondence in intraoral 3D scans. This enhanced model incorporated a geodesic map and a probability matrix to enhance the overall segmentation performance [36]. Additionally, Xu et al. [37] introduced a multi-stage CNN architecture for intraoral 3D scan segmentation. This approach involved training two independent CNNs to achieve teeth–gingiva separation and teeth labeling.

In the same vein, Zanjani et al. [38] applied PointNet architecture [39] to achieve semantic segmentation directly on intraoral 3D point clouds in order to segment the gingiva and individual teeth. Additionally, they employed an additional CNN as a discriminator head within an adversarial learning scheme to refine the teeth recognition task. The PointNet architecture was also adapted by Lian et al. [5] through the integration of graph-constrained learning modules. This modification aimed to facilitate the extraction of local features at multiple scale levels, specifically for the segmentation and labeling of teeth. Unlike Lian et al. [5] and Zanjani et al. [38], Tian et al. [40] added an extra step in the preprocessing in which intraoral 3D scans were initially encoded via a sparse voxel octree partitioning. Then, the encoded data was fed into a three-level hierarchical CNN for the segmentation, while a two-level hierarchical CNN was used for teeth labeling. Cui et al. [6] presented a different approach that consists of two main steps. Firstly, a dedicated CNN was employed for predicting 3D centroids as a means of teeth localization. Subsequently, a second CNN was applied individually on each tooth candidate to perform both tooth/gum separation and tooth type recognition. Similarly, Zanjani et al. [41] introduced a region proposal network (RPN) utilizing a Monte Carlo approach as an initial step to generate teeth detection candidates. Then, a modified Mask R-CNN architecture is applied to each generated tooth candidate for refining tooth instance segmentation. A lookup table based on the teeth centroids is finally used to assign labels to the previously detected teeth instances. Ma et al. [42] proposed an alternative deep neural network architecture designed for pre-detected teeth classification on the 3D point cloud of the scan. This architecture leverages adjacency similarity and relative position feature vectors to model the spatial relationship between teeth, thereby enhancing the labeling process. In their work, Zhao et al. [43] introduced a sequence of graph attentional convolution layers and a global structure network. These two networks aim to extract fine-grained local and global geometric features. Subsequently, these are combined to learn both the segmentation and the labeling tasks. Zhao et al. [44] proposed a graph convolutional network analyzing two distinct streams based on DGCNN [45]. The first stream of the network focuses on capturing the coarse structures of teeth using the 3D point cloud coordinates, while the second stream extracts distinct structural details by analyzing the

Table 1
Summary of studies proposing deep learning-based approaches for 3D dental segmentation.

Methods	Architectures	Resampling	Task	
			Segmentation	Labeling
Xu et al. [37]	Multi-stage	✓	✓	✓
Zanjani et al. [38]	End-to-end	✓	✓	✗
Zanjani et al. [41]	End-to-end	✗	✓	✓
Tian et al. [40]	Multi-stage	✓	✓	✓
Lian et al. [5]	End-to-end	✓	✓	✓
Zhang et al. [4]	End-to-end	✗	✓	✗
Sun et al. [34]	Multi-stage	✓	✓	✓
Sun et al. [36]	End-to-end	✓	✓	✓
Cui et al. [6]	Multi-stage	✓	✓	✓
Ma et al. [42]	End-to-end	✗	✗	✓
Zhao et al. [43]	End-to-end	✓	✓	✓
Zhao et al. [44]	End-to-end	✓	✓	✓
Qiu et al. [7]	Multi-stage	✓	✓	✗
Zhuang et al. [47]	Multi-stage	✓	✓	✓
Krenmayr et al. [46]	End-to-end	✓	✓	✓

normal vectors of the mesh. Recently, Krenmayr et al. [46] introduced DilatedToothSegNet (DTSegNet) which extends the DGCNN concept by employing dynamic edge convolution to capture highly informative local features. These features are enhanced by incorporating more distant features through multi-layer sampling from an expanding but sparse k-nearest-neighbor graph. This approach simulates an increasing receptive field while maintaining the network’s ability to learn semantically meaningful local features.

The Dental Arch (DArch) method was proposed by Qiu et al. [7] as an alternative to existing learning-based methods that require costly point-wise annotations. The DArch method utilizes weakly annotated data, which includes annotations for tooth centroids and only a few teeth per dental scan, for 3D tooth instance segmentation. It is composed of two stages, namely tooth centroid detection and tooth instance segmentation. The dental arch is generated in the first stage by employing Bezier curve regression, while in the second stage, a graph-based convolutional network (GCN) is utilized to refine the dental arch and achieve accurate tooth instance segmentation. This approach aims to reduce the annotation costs associated with learning-based methods.

Recently, Zhuang et al. [47] introduced a Robust Hybrid Learning (RHL) network that capitalizes on the benefits of both semantic and instance segmentation in a multi-task framework. In this approach, the predictions are performed on down-sampled point clouds resulting in the loss of many points along the teeth–gingiva boundary. Consequently, when the predicted labels are interpolated to the original models, the segmentation boundaries of the teeth lack smoothness [47].

Table 1 summarizes the aforementioned deep learning-based studies for intraoral 3D scan segmentation and labeling, emphasizing the key features such as architecture, data sampling, and addressed tasks.

2.3. Method positioning

Many conclusions can be drawn from the previously presented state-of-the-art works. The first is that geometry-based approaches often fail due to their poor robustness against complex tooth shapes and irregular teeth arrangement. A fully automatic and robust perception of intraoral 3D scans is needed to unlock the full potential of artificial intelligence in CAD systems.

Deep learning has unsurprisingly become the gold standard approach for segmenting and labeling teeth in intraoral 3D scans. Almost all recent studies on teeth localization, segmentation, or labeling tasks are based on deep learning architectures. However, the majority of these studies use a small number of scans, typically less than 100, implying potentially less than 50 patient use cases. The obtained results on such limited datasets should be taken cautiously as they can lead to overly optimistic evaluations and may not generalize on new cases.

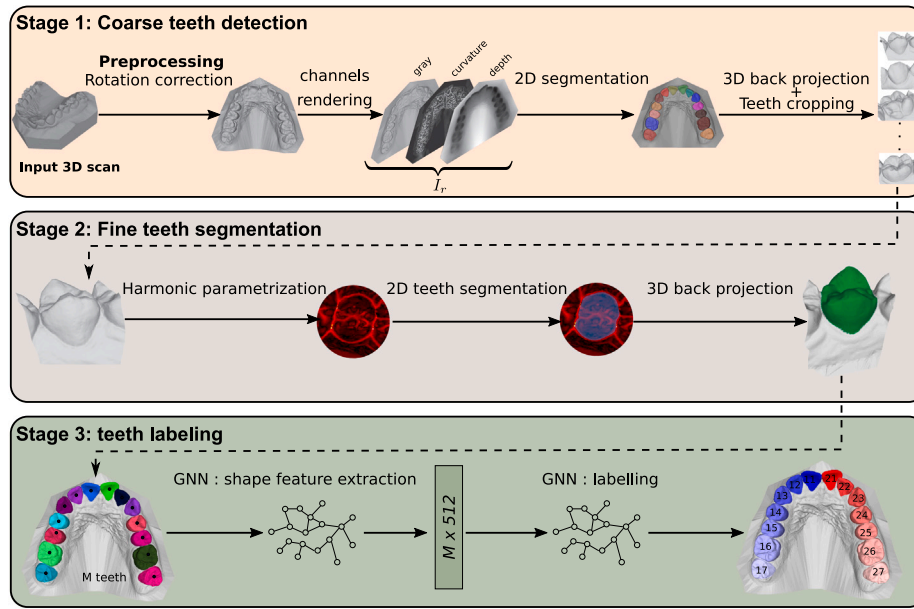


Fig. 1. The proposed segmentation pipeline consists of three main stages. 1- Coarse teeth detection: based on detecting teeth in the rendered three-channel 2D representation of the input scan using mask-RCNN. 2- Fine teeth segmentation in 2D harmonic parameter space to separate the crown from the gingiva and surrounding teeth. 3- Teeth labeling: To identify each segmented tooth in the input intraoral 3D scan based on GNNs.

We recently released a dataset that comprises data collected from 900 patients [8] and we also organized the 1st edition of the 3DTeethSeg scientific challenge in conjunction with MICCAI'22 [48]. Working with such a large dataset, encompassing diverse patient genders and ages significantly enhances the accuracy and robustness of the developed models and allows their systematic benchmarking.

Although popular deep learning segmentation algorithms such as Pointnet++ [39] or point cloud transformer [49] can produce good segmentation results on classic datasets objects, they encounter the problem of balancing between computational complexity and segmentation accuracy when applied to high-resolution intraoral 3D scans. To address this issue, we observe that the most widely adopted approach is to divide the input intraoral 3D scan into several patches to reduce the amount of computation while maintaining sufficient local features [6,41]. This is an intuitive scheme to allow the remaining components of the pipeline to focus only on the teeth segmentation and labeling tasks. In our approach, we propose handling the teeth detection process by adapting a 2D object detection baseline to a three-channel representation, including depth, gray-level, and curvature maps, which are automatically rendered for the input 3D scan. This approach benefits from the advances in 2D object detection solutions and it is computationally efficient since no severe down-sampling is required for the highly dense 3D scans. The teeth localization is done in a single pass on a batch of images to obtain coarse 2D masks but robust detection results. In addition, we handle teeth segmentation and labeling separately. Following the localization of each tooth, a straightforward mapping to 3D scans is done in order to crop each tooth candidate with a tight margin on its surroundings. The crown of the detected tooth is then separated from the gingiva and surrounding teeth using a semantic segmentation method (see Section 3.2).

Teeth labeling is the final process of our pipeline. To the best of our knowledge, Ma et al. [42] is the only study that specifically attempts to consider the spatial arrangement between adjacent teeth. We believe that, in addition to tooth shape and appearance, the spatial distribution of teeth relative to each other in the jaw is particularly informative on the type of teeth, especially in the case of damaged teeth. This strategy needs to be thoroughly investigated. In this article, we describe a new GNN that operates only on segmented teeth crowns. Graphs are suitable for spatially unstructured nodes, such as teeth in our case. Graphs also

allow to control message passing from neighboring nodes via an attention mechanism that can take as input the relative positioning of nodes. Furthermore, unlike regular MLP or fully connected layers, GNNs are versatile to varying numbers of nodes. This is particularly relevant when it comes to missing teeth cases as shown in the experimental Section 4.

To summarize, our contributions in this work can be outlined as follows:

- Streamlined 3D teeth detection and segmentation by a 2D image analysis formulation.
- Leveraging principal curvature and harmonic mapping to enhance teeth segmentation accuracy and achieve smooth teeth boundaries.
- A novel GNN architecture that incorporates both shape analysis and spatial arrangement for precise teeth labeling.
- Extended experimental validation of the proposed method on a publicly available benchmark, demonstrating the superiority of our approach.

3. Proposed approach

In this section, we present our approach for intraoral 3D scan segmentation and labeling. As described in the previous section, end-to-end classic 3D segmentation algorithms are not adapted to accurately segment high-resolution intraoral 3D scans. Therefore, we propose a coarse-to-fine multi-stage approach for segmentation and labeling. As depicted in Fig. 1, the full process is a pipeline starting with a coarse teeth detection, followed by a fine teeth crown segmentation, and finally the identification of the previously detected teeth. In this way, each task is based on the output of the previous step. Consequently, it is critical to ensure robust teeth detection at the initial stage of our pipeline since any false negative or false positive cannot be recovered later in the pipeline. The remainder of this section goes over each stage in detail.

3.1. Coarse teeth detection

The purpose of this step is to divide the input 3D intraoral scan into multiple patches, each corresponding to a tooth and its surrounding

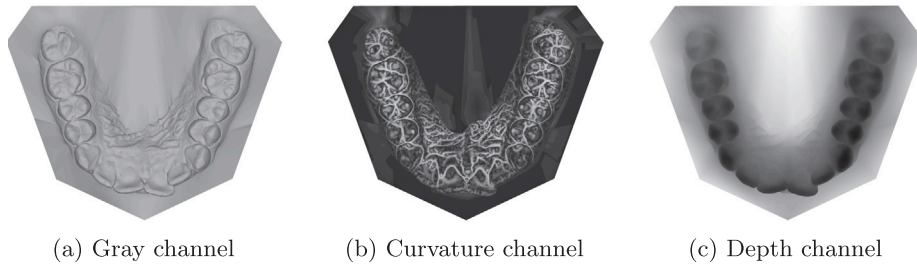


Fig. 2. Rendered channels for coarse teeth detection.

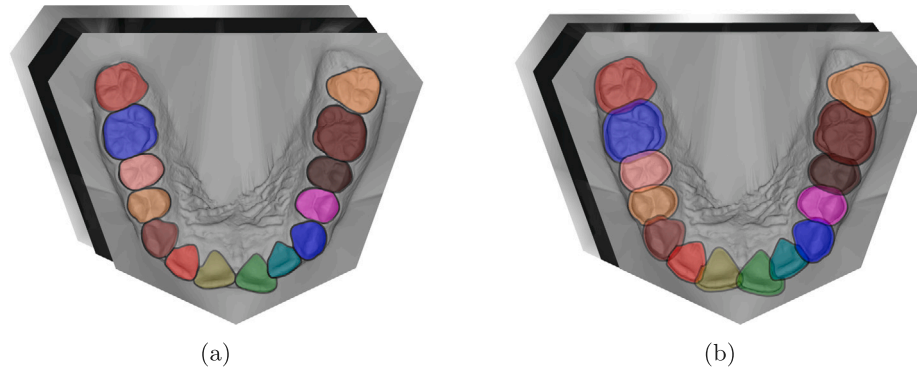


Fig. 3. Example of teeth localization in I_r images. **a:** Resulting segmentation teeth masks. **b:** Dilated segmentation teeth masks.

region. This process facilitates the subsequent fine segmentation step, which can then concentrate on each tooth separately.

Intraoral scans could be considered as a controlled scene and we can leverage the topology of the teeth surfaces and the pose of the dental arch to render synthetic images of the 3D scans optimal for 2D teeth detection. 2D object detection models are attractive in our case for their efficiency in terms of GPU memory usage and detection performance.

3.1.1. Preprocessing step

Depending on the used camera, intraoral 3D scans exhibit varying levels of noise, 3D point resolution, and initial 3D pose. Therefore, a preprocessing step is required to unify input scans. First, all degenerated and redundant mesh faces, as well as duplicated and irrelevant vertices, are deleted using a standard 3D analysis tool. The 3D scans are then automatically centered and rotated to fit the occlusal plane by predicting the oriented bounding box using principle component analysis applied to the 3D coordinates and normal vectors. These steps allow for a normalized occlusal viewpoint in which all teeth are visible.

3.1.2. Teeth candidate proposal

After preprocessing, the input scan in the occlusal plane is rendered to generate gray, curvature, and depth channels as illustrated in Fig. 2. The curvature channel is obtained by encoding the 3D curvature values in the vertex colors of the dental scan. The rendered channels are merged to obtain a three-channel image I_r , where all teeth in the dental scan are visible.

To detect teeth in I_r , we train a Mask R-CNN model. It is worth noting that variate of mask R-CNN model operating directly on 3D medical images exists, such as Jeong et al. [50]. However, to the best of our knowledge this model is not notorious for 3D point cloud analysis and few publications claiming this type method are more suitable for medical imaging modalities like MRI [50], making it less relevant for our use case.

Our training dataset is performed on the Teeth3DS dataset [8] as follows. We used PyTorch3D library as a 3D rendering engine to generate the three-channel of I_r for each intraoral 3D scan and the corresponding set of ground truth 2D masks defined on the rendered image for each

tooth separately. This first stage of our multi-stage pipeline focuses solely on detecting tooth instances. Assigning tooth labels (e.g., incisors, molars, etc.) to the detected teeth is handled separately in the final stage of the pipeline. Therefore, all teeth are treated as identical objects to be detected in this initial stage. During the training stage, we apply data augmentation by uniformly performing random vertical flips and rotations within a range of $[-10, 10]$ degrees on the I_r image.

This coarse teeth detection approach is particularly efficient regarding computation time and detection performance. Indeed, the 3D information encoded in I_r is sufficiently discriminative to distinguish not only between teeth and gum, but also between teeth instances, while remaining invariant to the 3D point cloud resolution of the scan. Another option would be to render several images for the same 3D scan under multiple 3D poses and use Mask R-CNN to detect the teeth in all of them. This approach would require an additional fusion step, which could be performed using the Non-maximum Suppression algorithm. We investigated this approach and did not find a significant improvement in terms of detection performance when compared to using only the rendered image corresponding to the occlusal plane.

After this step, we obtain a set of detected teeth 2D masks in I_r coordinate system. Each resulting mask is dilated to cover the surrounding area of the detected tooth instance (see Fig. 3), which can be coarsely located in the 3D scan using the same 3D/2D projection matrix that produced I_r .

3.2. Fine teeth segmentation

The fine teeth segmentation step consists of accurately separating the crown of each detected tooth from the gum as well as the surrounding teeth. We propose to perform the fine teeth segmentation task in the 2D harmonic parameter space. As shown in Fig. 4, the tooth mesh is mapped isomorphically to a 2D harmonic parameter space [51] as a planar 2D mesh.

The cropped tooth mesh is a non-closed 3D surface with only one boundary. Suppose the parameter formulation of surface $M \subset \mathbb{R}^3$ relative to the point (u, v) in a plane space $D \subset \mathbb{R}^2$ is:

$$r(u, v) = (x(u, v), y(u, v), z(u, v)) \quad (1)$$

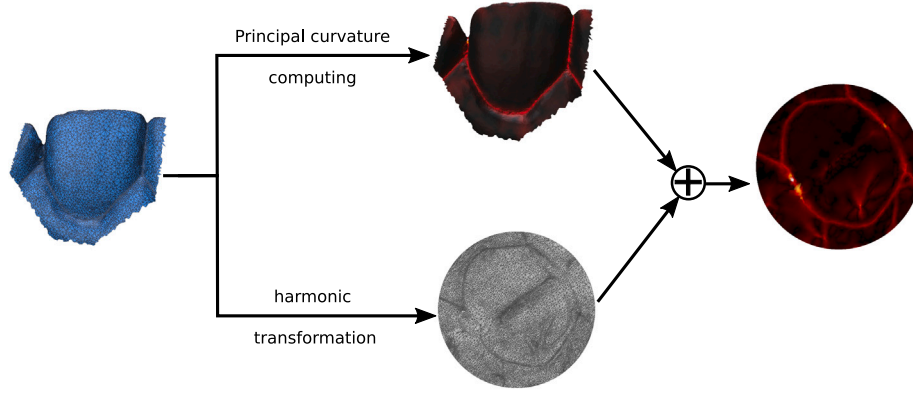


Fig. 4. Harmonic parametrization. The 3D tooth surface is mapped to a circular 2D plane resulting in a 2D mesh. The principal curvature is computed on the 3D tooth surface and encoded in the 2D mesh.

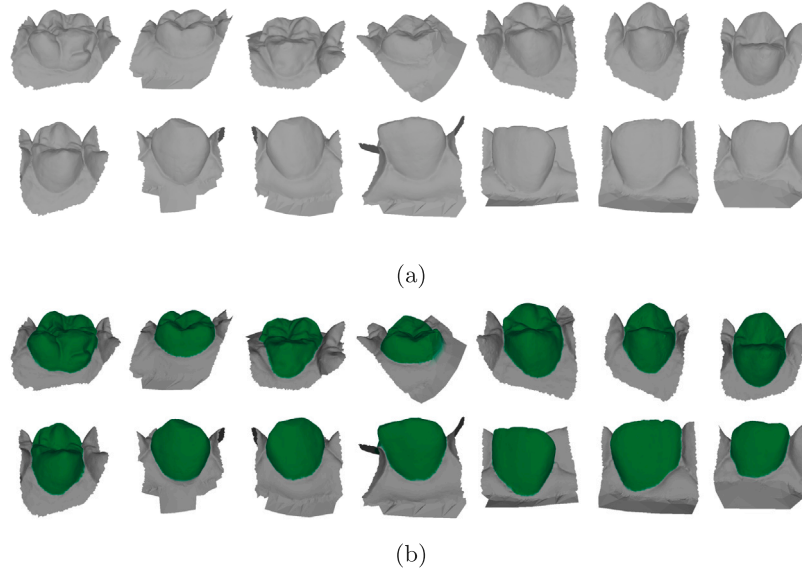


Fig. 5. Example of crown segmentation results obtained for different teeth.

We used the convex representation method for surface parametrization by fixing a circular polygon boundary ∂D of the plane D , that linearly maps the boundary ∂M of the tooth surface M to ∂D . Then, the coordinates (u, v) of the internal vertices of the plane D can be determined by energy minimization [4].

Moreover, the principal curvature is computed on the 3D mesh to provide information about the local shape of the tooth surface at each vertex. Then, we perform a min_max normalization on the principal curvature information before encoding it into the harmonic map (see Fig. 4).

Our tooth fine segmentation method is based on the Mask R-CNN algorithm: it takes as input the harmonic map with the curvature information and predicts the tooth segmentation mask. Similar to the training of the tooth localization model, we apply data augmentation by randomly performing vertical flips and uniformly rotating the harmonic map image within a range of $[-10, 10]$ degrees. The obtained segmentation mask is back-projected to the original cropped mesh to retrieve the tooth vertices. Each cropped tooth region undergoes individual segmentation, which can result in conflicting vertices at the boundaries between adjacent teeth. In such cases, a vertex might be assigned to two different teeth instances simultaneously. To ensure that each vertex is associated with only one tooth, any conflicting vertex is assigned to the tooth with the closest boundary. Fig. 5 shows an example of the segmentation results obtained for different teeth types.

3.3. Teeth labeling

The proposed teeth labeling approach is based on a GNNs that considers both tooth shape and spatial arrangement as clues. It involves two sub-networks. A first graph GNN-TA, which stands for Tooth Appearance GNN, and a second graph GNN-TL, standing for Teeth Labeling GNN. In the GNN-TA each node in the graph corresponds to the vertices of the tooth instance, and it is used to extract a meaningful representation from each tooth instance independently. In the second graph GNN-TL, each node represents a tooth instance. It is used to perform the teeth labeling by taking as input the computed GNN-TA representation of each tooth. By using GNNs in this two-step process, our approach can effectively leverage both tooth appearance and spatial arrangement to accurately label teeth. The pipeline for teeth labeling is illustrated in Fig. 6. Below, we provide a description of the key components involved in the GNN-TA and the GNN-TL.

3.3.1. GNN-TA

The objective of this graph is to focus only on the appearance of a single tooth, thus we explicitly filter both translation and rotation variations from each input tooth crown. Data centering is a common operation to filter translation variations. Filtering the rotation variation has been addressed in a number of works relating to 3D point cloud

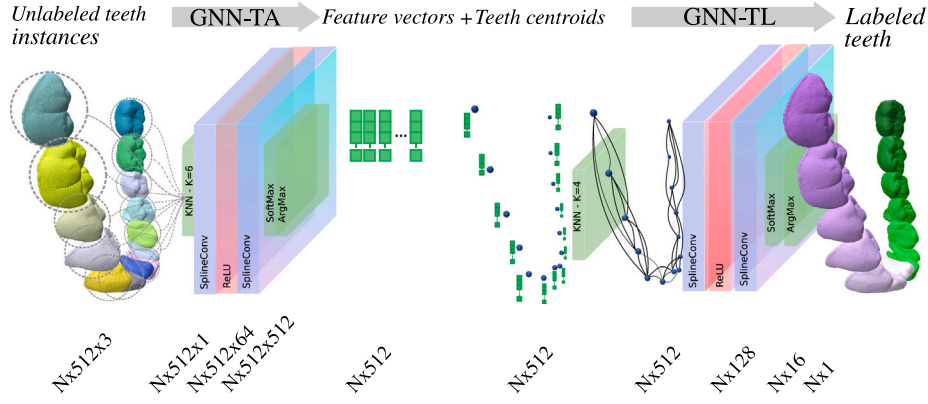


Fig. 6. Details of the teeth labeling pipeline, illustrating the GNN-TA and GNN-TL architectures. The feature vectors generated by the GNN-TA are mapped to the teeth centroids to serve as input nodes for the GNN-TL. The dimensions below indicate the tensor dimensions at each step of the pipeline, with N representing the number of previously detected teeth.

analysis, beginning with PointNet’s pioneer work, which proposed a “T-Net” to predict rotation transformations that can be applied to the input 3D coordinates as well as hidden spaces. In our approach, the input intraoral 3D scan is first centered, and the teeth crowns are then aligned on the occlusal plane in the preprocessing step. Afterward, we replaced the input vertices 3D coordinates by the Euclidean distance computed between the tooth centroid and all of the vertices so that GNN-TA remains fully rotation invariant. The adjacency matrix encoding the edges of GNN-TA is computed using a KNN algorithm with $K = 6$ neighbors. Also the GNN-TA has two graph convolution layers with a ReLU activation in between followed by a global max pooling that outputs the channel-wise maximum across the node dimension and yields a feature vector of 512 to characterize each tooth instance.

3.3.2. GNN-TL

The idea behind the GNN-TL graph is that it should benefit from the appearance representation extracted with the GNN-TA and allow a message passing from the neighboring teeth to obtain the final teeth classification.

Input graph. This graph is defined by $GNN-TL = (\mathcal{V}, \mathcal{E})$ with $\mathcal{V}$ being the set of nodes (*i.e.*, teeth instances) and $\mathcal{E}$ the set of edges between the nodes. Let $v_i \in \mathcal{V}$ denote a node and $e_{i,j} \in \mathcal{E}$ denote an edge pointing from v_j to v_i . The neighborhood of a node v_i is defined as $\mathcal{N}(v_i) = \{v_j \in \mathcal{V} | e_{i,j} \in \mathcal{E}\}$. In addition, we compute the Cartesian pseudo-coordinates $\mathbf{u}(i, j) \in [0, 1]^2$ for each directed edge $e_{i,j} \in \mathcal{E}$. Here, $\mathbf{u}(i, j)$ is two-dimensional because the jaw is already positioned on the occlusal plane, and therefore there is little interest in including the 3rd dimension as it would slow down the convolution computations. The pseudo-coordinates $\mathbf{u}(i, j)$ can be gathered in an adjacency matrix U , with normalized two-dimensional entries $u(i, j) \in [0, 1]^2$ if $e_{i,j} \in \mathcal{E}$ and 0 otherwise.

Input node features. The i th node v_i is initially represented at the input of the graph by the feature vector $\mathbf{x}_i \in \mathbb{R}^{512}$ which is already computed by applying GNN-TA on the i th segmented tooth crown. While the node attributes $\mathbf{x}_i$ are relative to each tooth separately, message passing between the nodes should be driven by the spatial arrangement of neighboring teeth. In this way, the classification of a given tooth is grounded on the aggregation of all teeth shape representations and their spatial arrangement.

Graph architecture. GNN-TL consists of 2 SplineCNN convolution layers [52]. SplineCNN convolution originally denoted by $(\mathbf{f} \star \mathbf{g})(i)$ is designed to analyze 3D shapes. It is computed as follows:

$$(\mathbf{f} \star \mathbf{g})(i) = \frac{1}{|\mathcal{N}(v_i)|} \sum_{l=1}^{512} \sum_{v_j \in \mathcal{N}(v_i)} \mathbf{x}_i(l) \cdot g_l(\mathbf{u}(i, j))$$

where $|\mathcal{N}(v_i)|$ is the cardinal of v_i neighbors, $\mathbf{x}_i(l)$ indicates the l th dimension in the $\mathbf{x}_i$ feature vector, and g_l is the kernel function with a kernel size of 4 and a max operator as aggregation.

3.3.3. Data augmentation

Recent studies have highlighted the significance of data augmentation techniques in enhancing results for various medical imaging tasks [53]. For example, Farook et al. [54] shows that extruding scan vertices along the normal axis improves classification of upper and lower jaws. In this section, we present the data augmentation technique employed to enhance the predictions of our labeling model.

One advantage of using graphs for teeth labeling is the ability to handle a variable number of nodes. This includes, for instance, the challenging case of missing teeth. As the training data should be representative of this kind of configuration, we perform different types of data augmentation. The most common method involves applying uniform random shifts within a range of $[-20, 20]$ mm and small rotations uniformly within a range of $[-5, 5]$ degrees along the three axes on the input scan. For the GNN-TA graph, we introduce a small jitter on the teeth centroid coordinates within a range of $[-t_s, t_s]$, where t_s is the tooth size. This jitter creates variations in the input features, which are defined by the Euclidean distances between the teeth surface points and their corresponding centroids. Additionally, to simulate missing teeth in the training dataset and ensure a balanced distribution of cases with varying numbers of teeth in the arch, we intentionally remove teeth from the inputs of the GNN-TA and GNN-TL graphs based on computed statistics of missing teeth. This is achieved by implementing a random drop nodes function that randomly removes a subset of teeth. This operation is applied exclusively to scans containing 14 teeth.

4. Experiments and results

This section aims to evaluate the performance of the proposed method on a public benchmark. Our evaluation strategy consists of two levels. First, we intrinsically evaluate the performances of the three different tasks involved in our approach: coarse teeth detection, fine teeth segmentation, and teeth labeling. Then, our method is globally evaluated and compared with state-of-the-art methods. We note that all experiments presented in this section are performed on a computer equipped with an Intel(R) Xeon(R) V3 CPU, an Nvidia Titan V GPU (12 Gb), and 64 GB RAM. Both Mask R-CNN models for localization and segmentation are trained using the Stochastic Gradient Descent (SGD) optimizer. The initial learning rate is set to 0.00025, with a decay factor of 0.0001. Training is carried out for 200 epochs, with a batch size of 4. Our labeling graph model was trained for 500 epochs using the following hyperparameters: a batch size of 32, a learning rate of 0.001, and a weight decay of 0.001 for regularization. The

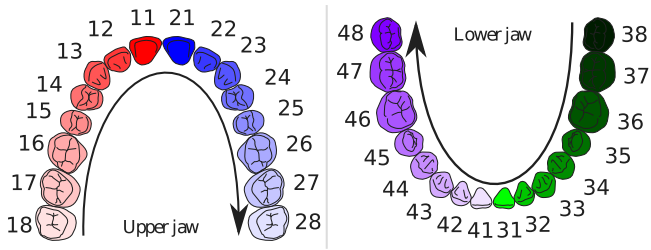


Fig. 7. FDI World Dental Federation notation [8] and the corresponding color coding.

architecture comprised two GCN layers with convolutional filter sizes of 64 and 512, respectively, followed by two SplineConv layers with a kernel size of 4. To transition from GNN-TA to GNN-TL, we applied global max pooling to reduce the feature dimension from (512, 512) to 512 per node (tooth). In the GNN-TA, we used a k-nearest neighbors (knn graph) approach with $k = 6$. In contrast, the GNN-TL employed a knn graph with $k = 4$ and incorporated spherical coordinates in the edge attributes for learning by the SplineConv layers. We optimized the hyper-parameters of our models using the Optuna tool, testing a range of parameter values. The primary selection criterion was validation performance, which varied depending on the model (i.e., Mask R-CNN, GNN models).

In the remainder of this section, we start by presenting the dataset used for our experiments. Then, we detail the conducted experiments and the obtained results to intrinsically evaluate each task in our approach. Finally, we quantitatively and qualitatively assess the overall approach.

4.1. Evaluation dataset

The proposed method is evaluated on the Teeth3DS benchmark [8] consisting of 1800 intraoral 3D scans collected from 900 patients and covering the upper and lower jaws separately. The dataset is originally split into 1200 scans for training and the remaining 600 scans for testing. The scans selection was carried out to maintain a consistent training/testing proportion at the level of tooth labels, ensuring that the data remains balanced. Teeth distributions are consistent between the upper and lower jaws. For instance, 10.7% of upper jaws and 10.3% of lower jaws have 13 teeth. The reader can refer to Ben-Hamadou et al. [8] for more in-depth details about the Teeth3DS dataset.

It is worth noting that Teeth3DS is the first publically available dataset for intraoral 3D scan segmentation and labeling. As shown in Fig. 7, the tooth numbering system is based on the FDI World Dental Federation notation. The used color coding is consistent with our reported labeling results.

4.2. Tasks intrinsic evaluation

We started the experiments by validating that 100% of teeth were visible after applying the preprocessing PCA transformation to adjust the occlusal plane of the 3D scans. This ensures that all teeth are thus potentially detectable from the generated rendered image I_r .

4.2.1. Teeth localization evaluation

Teeth localization is a critical step in our method since it occurs at the very beginning of our process. Indeed, any missed or false detection could not be recovered later on in our pipeline. As the detection is performed on the rendered image I_r , we considered as a metric the standard Intersection Over Union (IoU), which measures the overlap between the detected and ground truth 2D bounding boxes.

A detection is considered as true positive (TP) if it meets two criteria: (i) its detection confidence score surpasses a predefined threshold, and (ii) the associated predicted bounding box achieves an IoU

Table 2

Average precision obtained for different IoU thresholds σ .

	$\sigma = 90$	$\sigma = 85$	$\sigma = 75$	$\sigma = 50$
AP	90.8	95.9	98.1	99.7

Table 3

Individual teeth segmentation performances using PointNet++, Kpconv and Harmonic mapping.

	Kpconv	PointNet++	Ours
F1-score	0.94	0.95	0.98

threshold σ with at least one ground-truth bounding box. On the other hand, a false positive (FP) occurs when the confidence score is above the threshold, but the predicted bounding box does not match any ground truth bounding box. Finally, a false negative (FN) pertains to a ground truth bounding box that lacks a corresponding detection, indicating that its highest IoU score is lower than σ .

Table 2 illustrates the obtained average precision (AP) values for teeth detection on the rendered image I_r for different IoU thresholds σ . These results highlight the accuracy of our proposed teeth detection method, even though it relies on the rendered image I_r instead of the complete 3D point cloud. To optimize accuracy while minimizing the false negative (FN) rate, we consistently set the confidence threshold to 0.97 for the remaining experiments.

4.2.2. Teeth segmentation evaluation

In this section, we intrinsically evaluate the tooth segmentation process which consists of separating the tooth crown from its surrounding teeth and the gingiva. We used the average F1-score overall tooth 3D vertices to evaluate the accuracy of the segmentation process. Our segmentation method is compared with two well-known 3D point cloud segmentation baselines: PointNet++ [39] and Kpconv [55]. It is important to note that these methods were trained on the same tooth crop dataset as our segmentation method to ensure a fair comparison. The obtained quantitative segmentation results are summarized in Table 3 where our method outperforms Kpconv and PointNet++ by 0.04 and 0.03, respectively. We additionally provide some visual examples of segmentation results in Fig. 8 where the first column presents the input coarsely detected tooth mesh, columns 2, 3, and 4 present the input mesh representation in harmonic space (HS), 2D segmentation results in HS and its back projection to 3D space (our results). Columns 5 and 6 in Fig. 8 present the visual segmentation results for PointNet++ and Kpconv methods. We may note that the segmentation results look similar globally among the compared methods. However, the subtle differences become particularly evident at the borders of the segmented teeth, where our method achieves significantly smoother and more accurate teeth borders compared to PointNet++ and Kpconv. These observations are consistent with the obtained quantitative results, demonstrating the effectiveness of considering locally applied harmonic space for segmentation rather than performing the segmentation directly on the 3D space.

4.2.3. Teeth labeling evaluation

Teeth labeling is the final process in our pipeline, where each detected and segmented tooth is assigned its corresponding FDI teeth class. The labeling performance was evaluated using the F1-score metric. Table 4 presents the obtained scores with a detailed per class and upper and lower jaws breakdown. We can notice that the best performances are obtained for the teeth classes that are the most present 3D intraoral scans, such as incisors, a slightly lower performances are general observed for teeth that are less present, such as wisdom teeth (i.e., labels 38, 18, 28, and 48).

To further analyze our labeling method, especially the level of confusion among different teeth classes, we compute confusion matrices for

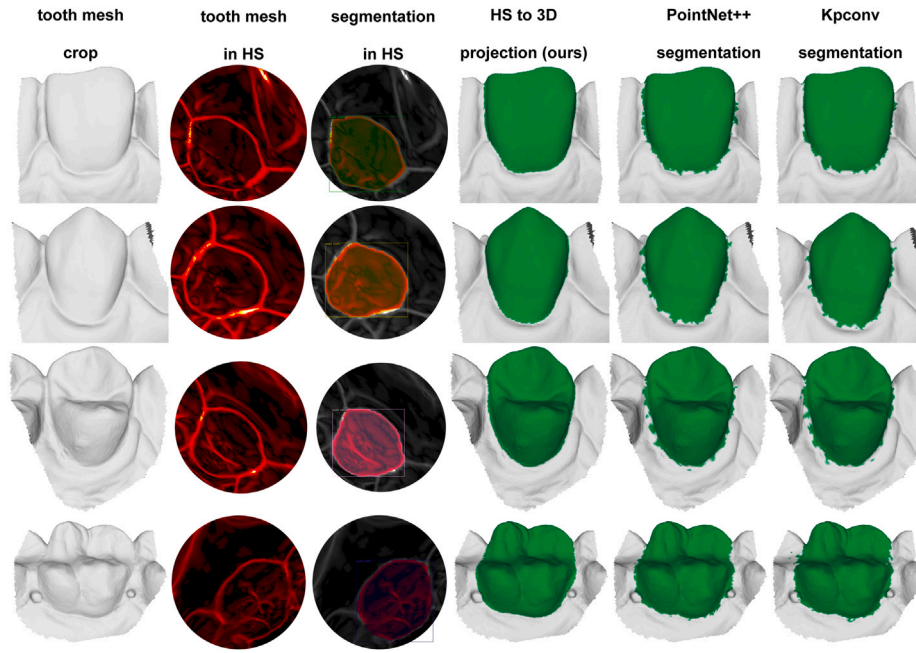


Fig. 8. Visual segmentation results for different tooth types (rows). From left to right: input mesh, tooth mesh representation in harmonic space (HS), 2D segmentation results in harmonic space, our segmentation results (back projection of HS segmentation to 3D space), PointNet++ segmentation results, Kpconv segmentation results.

Table 4

F1-score labeling performances for different teeth classes for upper and lower jaws. The support column indicates the number of teeth instances within each FDI class.

Upper			Lower		
FDI class	F1-score	Support	FDI class	F1-score	Support
18	0.92	13	38	0.93	15
17	0.94	201	37	0.94	223
16	0.96	298	36	0.91	295
15	0.95	288	35	0.96	291
14	0.97	292	34	0.95	292
13	0.98	269	33	0.97	293
12	0.97	297	32	0.97	299
11	0.97	300	31	0.95	296
21	1.00	300	41	0.96	299
22	0.98	298	42	0.95	299
23	0.99	274	43	0.96	296
24	0.97	297	44	0.95	297
25	0.97	286	45	0.96	289
26	0.94	300	46	0.94	292
27	0.94	211	47	0.96	218
28	1.00	11	48	0.93	14

both upper and lower jaws (see Fig. 9). As expected, we observe high values approaching 1 along the diagonal of both matrices, indicating accurate classification. The main sources of confusion are typically with the immediate adjacent teeth but rarely extend beyond that.

4.3. Global evaluation

4.3.1. Quantitative evaluation

To assess the performance of 3D segmentation and labeling methods for intraoral scans we use the standard semantic segmentation evaluation metrics: Overall Accuracy (OA) and Dice Score (DSC). Additionally, we considered the three metrics from the MICCAI 3DTeethSeg Challenge [48], which are defined specifically to evaluate the teeth localization, segmentation, and labeling tasks. We briefly introduce these metrics.

Teeth Localization Accuracy (TLA): evaluates the 3D teeth localization performance which consists of an exponential negative mean of

normalized euclidean distance between ground truth (GT) 3D teeth centroid and the closest localized teeth centroid. The euclidean distance is normalized by the size of the corresponding GT tooth where the tooth size is defined by the diameter of the bounding sphere. In case of no centroid (e.g., algorithm crashes or missing output for a given scan) a nominal penalty of 5 per GT tooth is given. This corresponds to a distance of 5 times the actual GT tooth size. As the number of teeth per patient may be variable, here the mean is computed over all gathered GT Teeth in the test dataset.

Teeth Segmentation Accuracy (TSA): evaluates the global teeth segmentation accuracy as the average F1-score over all instances of segmented teeth point clouds.

Teeth Identification Rate (TIR): evaluates the teeth labeling task. It is computed as the percentage of true labeling cases relative to all GT teeth labels. Given GT tooth, a true classification is achieved when the closest detected tooth centroid is located at a distance less than the bounding sphere radius and is labeled with the correct label. If a tooth is not detected or mislabeled, the cost associated with the false negative detection is high, as this task is very important in dentistry. This metric, in association of TLA, allow to assess the identification and localization capabilities of the approaches.

We note that all evaluation metrics used in our experiments originally range from 0 to 1, where 1 is the best value. For clarity, the obtained values for these metrics are presented as percentages.

To evaluate our proposed method, we conducted a comparative analysis against two generic methods (PointNet++, Kpconv) and four specialized methods: DTsegNet, RHL, MeshSegNet, and TSegNet. The performance of each method was assessed using the aforementioned metrics. The results are summarized in Table 5, where the highest values are highlighted in bold, demonstrating that our method consistently outperforms the other methods across all evaluation metrics. Specifically, it achieves improvements of 1.11%, 1.82%, 2.51%, 3.24%, 5.11% and, 4.37% in teeth localization (TLA) compared to RHL, PointNet++, Kpconv, TSegNet, MeshSegNet, and DTsegNet, respectively. Moreover, our method exhibits superior performance in teeth segmentation (TSA) compared to all methods. This trend is further supported by the evaluation of generic 3D segmentation metrics: DSC and OA.

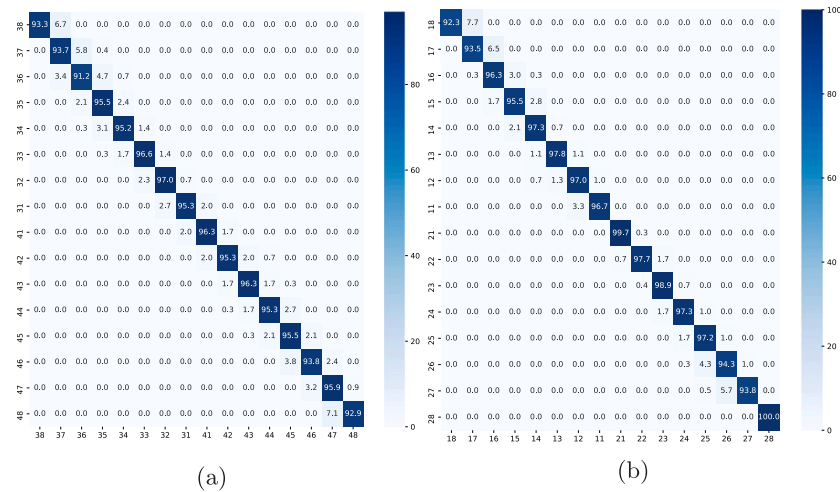


Fig. 9. Confusion matrices. (a) and (b) present confusion matrices for lower and upper jaws, respectively. The rows represent the true classes, and the columns represent the predicted classes. The values in each cell of the matrix indicate the percentage of instances that belong to a particular true class and were predicted as a particular predicted class.

Table 5

Comparisons with state-of-the-art methods on TLA, TSA, TIR, standard 3D segmentation evaluation metrics (DSC, OA), and running-time T in seconds.

	Methods	TLA $\uparrow$	TSA $\uparrow$	TIR $\uparrow$	DCS $\uparrow$	OA $\uparrow$	T ^a $\downarrow$
Generic methods	PointNet++	96.63	93.15	96.39	89.36	88.59	0.25
	Kpconv	95.94	92.68	96.77	88.93	88.22	0.61
Specialized methods	TSegNet	95.21	96.86	92.95	91.10	90.36	0.9
	MeshSegNet	93.34	87.02	83.19	87.51	87.88	3.1
	RHL	97.34	97.07	95.67	92.64	92.31	1.4
	DTSegNet	94.08	94.37	93.54	91.27	90.87	1.1
	Ours	98.45	98.17	97.61	95.10	94.30	5.3

^a Experiments are performed using an NVIDIA Titan V GPU.

As shown in the last column of [Table 5](#), our method takes approximately 5 s for segmentation and labeling. It is slower compared to other methods because it processes the full scan resolution and involves multiple stages, including harmonic mapping, curvature computation, and rendering.

To gain deeper insights into the overall performances reported in [Table 5](#), we conducted a detailed analysis of the Dice scores calculated on all scan vertices. This analysis involves a statistical breakdown based on different teeth categories with similar shapes, specifically incisors, canines, premolars, and molars. Also, we included gingiva as an additional shape in this breakdown, as it provides a relevant indicator for assessing the accuracy of teeth–gingiva separation. The resulting box plots of the Dice scores are depicted in [Fig. 10](#). We can observe that our method outperforms state-of-the-art methods in terms of mean and median for both gingiva and the different teeth categories. Furthermore, our method leads to smaller box plots, indicating smaller standard deviation values. We can also observe that our method has fewer outliers (black circles) than the other methods. It is worth mentioning that outlier samples with a Dice score of zero indicate undetected teeth. Additionally, when comparing the box plots for different methods on the gingiva region, we can observe that our method achieves a more accurate separation between teeth and gingiva surfaces than the state-of-the-art methods.

Finally, we observe a very comparable Dice score statistics across different tooth categories, with significantly higher scores for gum segmentation. This is due to the large number of gum vertices and the fact that segmentation errors are primarily located at the gum-tooth borders.

4.3.2. Qualitative evaluation

We present qualitative results for easy, medium, and hard cases in [Fig. 11](#), [12](#) and [13](#), respectively. All the cases presented are selected

from the test subset. Easy cases are free of artifacts in the gingiva surface and show no significant abnormalities in tooth shape or arrangement. Medium cases exhibit more artifacts and reconstruction errors in the gingiva area. Finally, hard cases contain significant abnormalities in gingiva, tooth shape, and tooth arrangement. In [Fig. 11](#), [12](#) and [13](#), black arrows indicate undetected teeth, while labeling errors and segmentation errors are marked with dotted and solid ellipses, respectively.

In general, the segmentation results produced by our method better match the ground truth. We notice that PointNet++, Kpconv, MeshSegNet and, DTSegNet usually produce lots of artifacts in the tooth boundary areas. Also, we observe that detection and segmentation errors are more prevalent in the easy and medium cases. However, our method outperforms other methods in labeling accuracy, particularly in the hard cases, demonstrating the effectiveness of our GNN-based labeling model.

We observe that the quantitative comparison produced in [Table 5](#) is consistent with qualitative results shown in [Figs. 11–13](#), which further demonstrates the effectiveness and efficiency of our method for tooth segmentation and labeling of intraoral 3D scans.

Although our method achieved prominent segmentation results and outperforms state-of-the-art methods, it does have some limitations that we acknowledge. For instance, in specific cases, such as wisdom or rudimentary teeth, our method may encounter challenges in teeth detection or produce incomplete tooth segmentation. We next present a few failure cases and provide insight into how they could be tackled in future work.

[Fig. 14](#) illustrates a few representative failure cases. For instance, our method failed to detect and segment the wisdom teeth in [Fig. 14\(a\)](#) as well as the rudimentary teeth in [Fig. 14\(b\)](#). It should be noted that rudimentary teeth are generally difficult to detect due to their small visible surface. Another example of failure is presented in [Fig. 14\(c\)](#), where our method detected the buccal and lingual premolar cusps as separate teeth instances. These failures are today attributed to the limited availability of samples representing these specific cases in the training dataset but could be, in the future, detected with statistical teeth shape models and/or statistical priors on the relative positions of the individual teeth. In this particular case, the individual shape of each detected tooth should be easily detected as an outlier, as well as their high proximity.

Furthermore, [Fig. 14\(d\)](#), [14\(e\)](#) and [14\(f\)](#) showcase visual results for three additional cases (not included in the Teeth3DS dataset): partial scan, scan with an abutment, and appliance cases, respectively. The obtained results reveal that our method lacks robustness when handling these particular cases. Our approach is designed to work with full arch

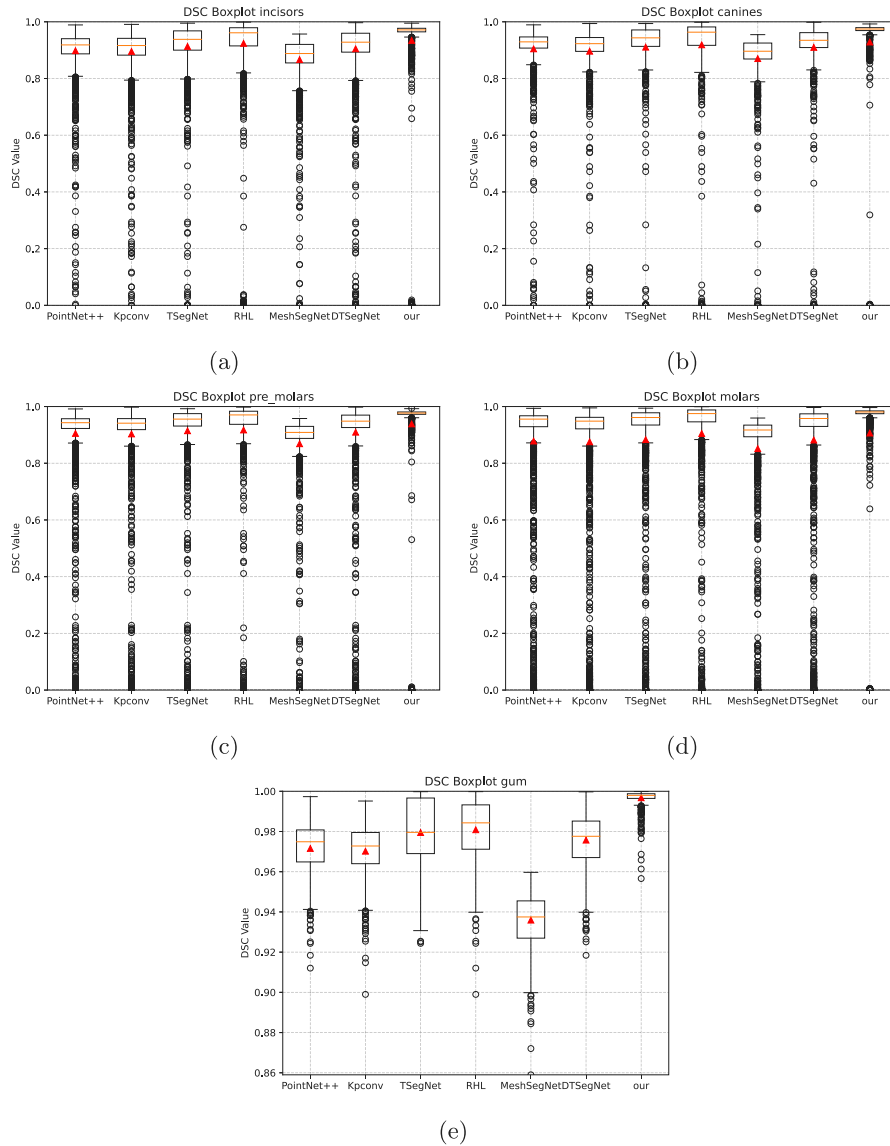


Fig. 10. Overall performance: Box plots comparing our method with state-of-the-art methods on the Dice score metric for different teeth shape classes (incisors 10(a), canines 10(b), premolars 10(c), molars 10(d)) and gum 10(e). The plots also show the mean (red triangle) and median (orange line) values of each measure.

scans with potentially missing teeth. However, it is not able today to provide correct teeth labeling for a partial scan with an arbitrary field of view of the arch. Also, our method is not able to correctly identify abutments as teeth and the teeth segmentation accuracy drops when appliances are present. The decrease in performance in these cases can be attributed to the absence of similar samples in the training dataset. Different augmentation strategies could be envisioned to improve the robustness to such cases.

4.3.3. Ablation study

I_r channels: as explained previously, the teeth localization process is based on three input channels: curvature, gray, and depth. In this experiment, we evaluate the impact of each channel on the localization performance. Table 6 summarizes the AP values for various σ values (see Section 4.2.1) and input channels. The obtained results illustrate that the optimal AP values for teeth detection are achieved through the combination of the three channels: gray, curvature, and depth. Fig. 15 shows example of activation maps obtained for our detection model, trained on different combinations of gray, curvature, and depth input channels. We used the Grad-CAM method [56] to generate the

activation maps highlighting the image regions contributing the most to teeth detection from rendering images. We observe that the activation maps predominantly focus on similar teeth regions and are very similar across different models, with only slight differences. For example, the depth-based model typically focuses on teeth regions with significant depth variation, whereas the curvature-based model shows more intense activation in regions with pronounced curvatures. We believe that each input channel contributes to the improvement of the teeth detection step, in line with the quantitative results in Table 6.

GNN-TA and GNN-TL complementarity: The proposed labeling pipeline is designed to consider both tooth appearance and spatial arrangement using GNN-TA and GNN-TL, respectively. We study the influence of each component on the labeling results. Table 7 summarizes the TIR values obtained using different labeling configurations. In the GNN-TA configuration, we used only the appearance graph architecture, where each node is followed by a fully connected layer for teeth classification. For the GNN-TL configuration, we used only the localization graph without appearance feature vector in the input of each node. The GNN-TA+TL configuration includes the full labeling model, comprising the appearance graph followed by the localization graph as described in Section 3.3.1. In Fig. 16, we present examples of

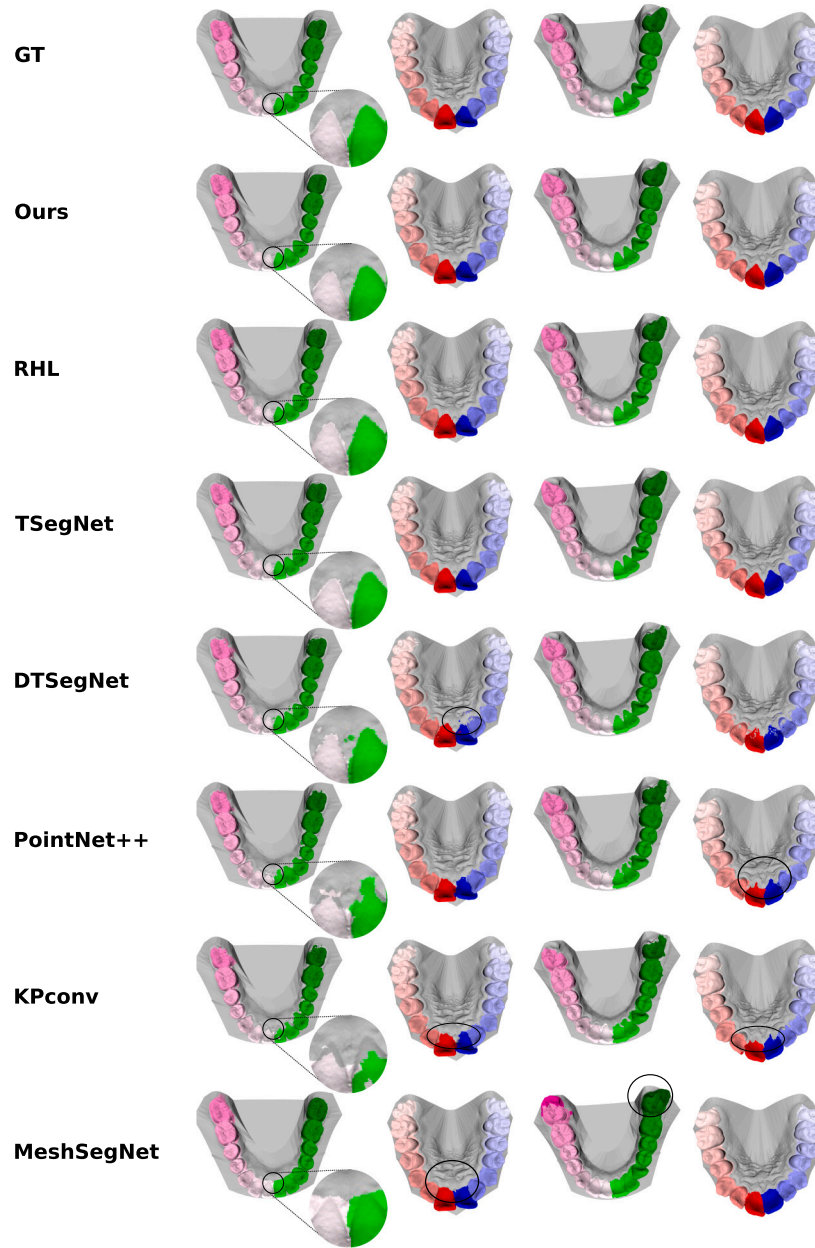


Fig. 11. Visual comparison of segmentation and labeling results for easy cases obtained by different methods. Each column corresponds to an example of either the upper or lower jaw.

Table 6

Average precision obtained for different IoU thresholds σ and input channels (curvature, gray, and depth).

	Curvature	Gray	Depth	$AP_{\sigma=90}$	$AP_{\sigma=85}$	$AP_{\sigma=75}$	$AP_{\sigma=50}$
Input channels	✓			88.7	93.9	96.8	97.7
		✓		87.0	91.8	95.4	96.9
			✓	88.1	94.2	96.8	97.2
	✓		✓	89.5	94.9	97.4	98.5
	✓	✓	✓	90.8	95.9	98.1	99.7

labeling results from these three configurations alongside the ground truth labels. The results in Table 7 show clearly that the appearance of a tooth (GNN-TA only) is sub-optimal for an accurate labeling in our setup (TIR equals 36.48%). This low performance could be explained by the fact that the appearance is informative about the category of a tooth (i.e., incisors, premolars, molars, canines), as clearly shown in the first row of Fig. 16. However, when it comes to differentiate between incisors, for instance between tooth 41, 42, 31, and 32, it becomes

insufficient. On the other hand, the obtained results for GNN-TL only, demonstrates that the spatial arrangement of teeth, here we used the centroid of each tooth as a spacial information, provides relevant features for labeling. Indeed, we obtained TIR equals 87.07% for GNN-TL only (see rows 2 in Fig. 16). Finally, when GNN-TA and GNN-TL are combined (GNN-TA+TL), we obtain the best performance, TIR equals 97.61% (see third row in Fig. 16), demonstrating the benefits of leveraging both GNNs in our design.

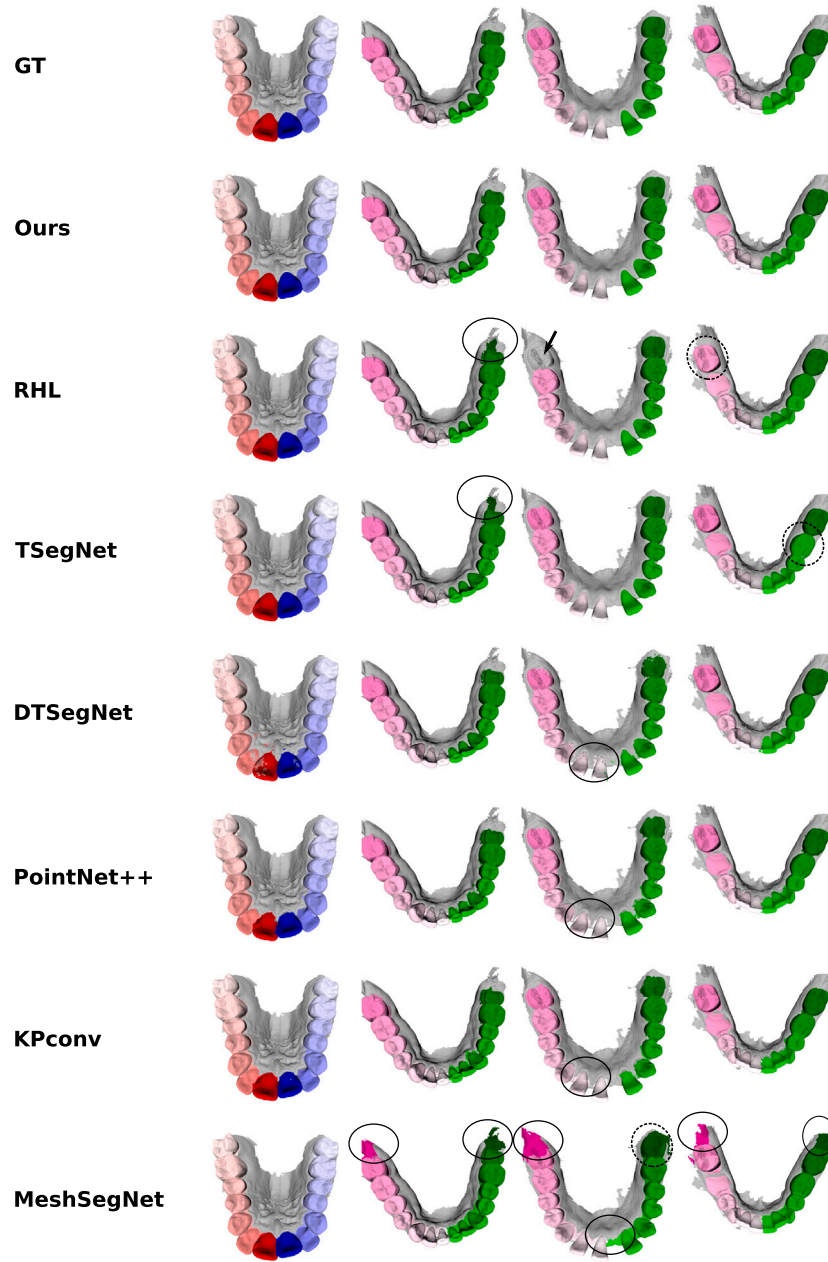


Fig. 12. Visual comparison of segmentation and labeling results for medium cases obtained by different methods. Each column corresponds to an example of either the upper or lower jaw. Black arrows denote undetected teeth, while labeling errors are marked with a dotted ellipse. Segmentation errors are highlighted with a solid ellipse.

Table 7

The obtained TIR values with different labeling configurations : GNN-TA, GNN-TL, and GNN-TA+TL.

	GNN-TA	GNN-TL	GNN-TA+TL
TIR	36.48	87.07	97.61

Labeling data augmentation: To assess the effectiveness of our data augmentation process, particularly in missing teeth cases, we computed the DSC metrics presented in Fig. 17. A general trend observed across all methods is a decrease in performance when teeth are missing, although the best performances are achieved with scans containing 14 teeth (missing 2 teeth). However, it should be noted that this case is the most represented in the dataset. Compared to the other methods,

our method consistently exhibits higher DSC values in all missing teeth cases. This experiment also highlights how the DA enhanced the labeling results, particularly in the different cases of missing teeth.

4.4. General discussion

Multi-stage vs. end-to-End design: In general, a multi-stage design offers several advantages over an end-to-end design. We particularly believe that our multi-stage approach allows for the specialization and optimization of each component individually, providing greater control over each step of the pipeline. For example, in the first step of our pipeline, teeth instance detection is treated as a relatively simple task, focusing solely on the localization of teeth without needing to resolve the complexities of teeth labeling. By simplifying the early stages, we ensure a more accurate and efficient process as the pipeline progresses to more challenging tasks (e.g., segmentation and labeling).

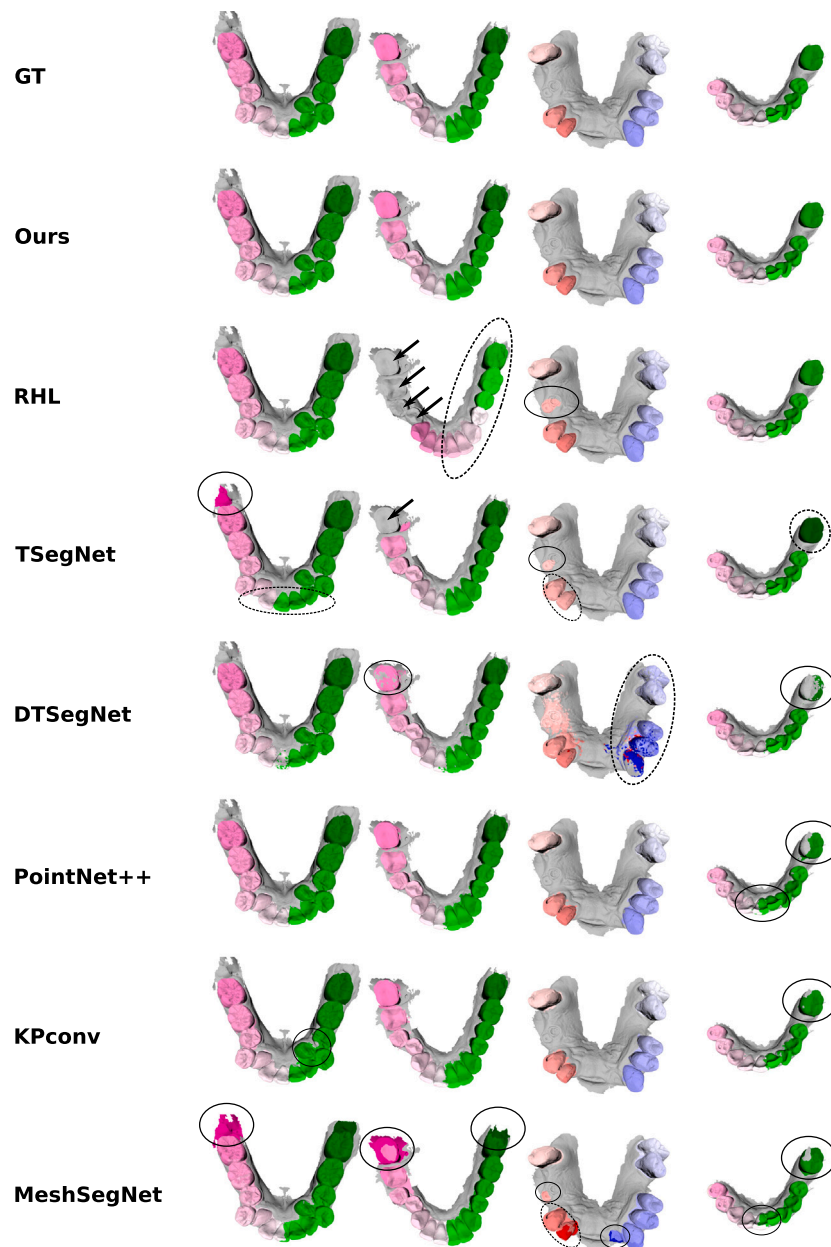


Fig. 13. Visual comparison of segmentation and labeling results for hard cases obtained by different methods. Each column corresponds to an example of either the upper or lower jaw. Black arrows denote undetected teeth, while labeling errors are marked with a dotted ellipse. Segmentation errors are highlighted with a solid ellipse.

Similarly, the teeth labeling step benefits from the output of the teeth instance segmentation, allowing it to focus exclusively on the shape and spatial arrangement of the already segmented teeth. This strategy reduces the complexity of labeling by building on the results of the previous stage. Additionally, a multi-stage approach enables the design of specific data augmentation techniques for each component of the pipeline. For example, the data augmentation strategy developed for training the teeth labeling model is particularly effective, as it simulates variations in teeth location and the presence of missing teeth, thereby enhancing the model's robustness and generalization capabilities.

On the other hand, one significant limitation of end-to-end models is their inability to effectively handle high-resolution input 3D scans. In such cases, end-to-end methods require a down-sampling step to reduce the point cloud to a fixed number of points (e.g., 8000, 16,000, 32,000) based on the machine's limitations. This down-sampling process leads to the loss of numerous vertices along the teeth-gingiva boundary.

Consequently, when the predicted labels are interpolated back onto the original models, the segmentation boundaries of the teeth often lack smoothness. In contrast, our approach is less sensitive to higher resolutions and offers the advantage of segmenting each tooth individually, preserving the maximum resolution, eliminating the need for label interpolation, and preserving the integrity of the segmentation boundaries.

Method Generalization: Our method does not rely on any assumptions about anomalies or treatments present in the 3D scans, whether in the training or test data. The Teeth3DS dataset includes a wide range of variability, as the scans come from various dental practices (orthodontists, surgeons, dental labs, etc.). We believe that our method is generalizable across a broad range of variations, provided it is trained on a representative training set. Indeed, we observed very few inaccuracies when processing intraoral 3D scans from similar providers (e.g., orthodontists, dental surgeons). However, failure cases can occur when

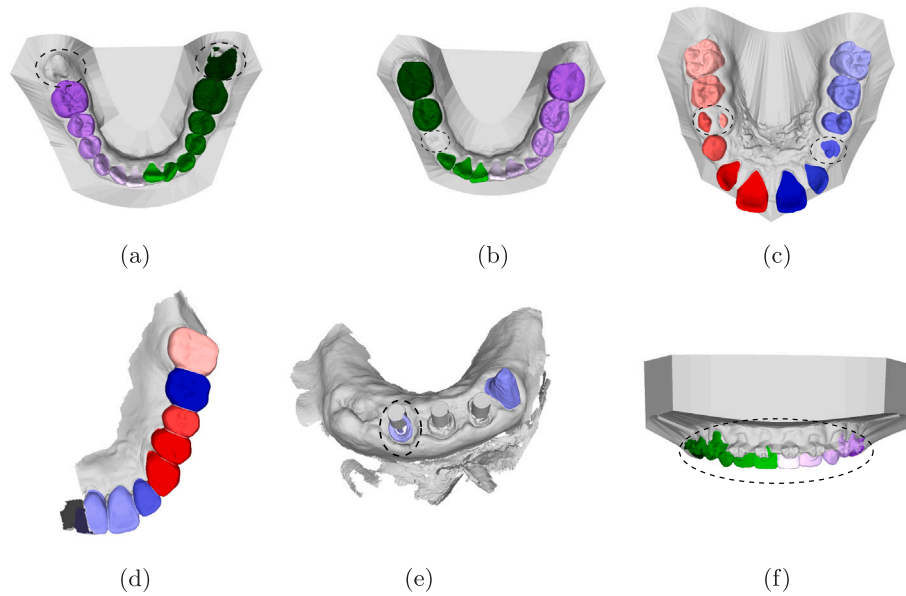


Fig. 14. Failed segmentation cases. (a) The wisdom teeth with abnormal shapes failed to be detected and segmented. (b) and (c): The rudimentary tooth is hard to detect and is not accurately segmented. (d): The labeling process is unsuccessful when dealing with partial scan cases. (e): Abutments may be mistakenly identified as teeth. (f): Teeth are not accurately segmented in cases where an appliance exists.

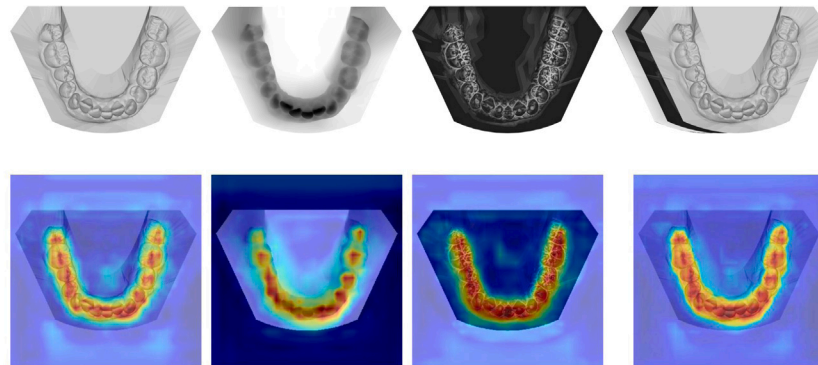


Fig. 15. Activation maps obtained from our teeth detection model trained on different channel combinations. From left to right: gray, depth, curvature, and all channels combined.

the 3D scans significantly deviate from the distribution of Teeth3DS, as illustrated in Figs. 14(d), 14(e), and 14(f). Self-supervised learning could be an alternative approach for training parts of the algorithm, potentially enhancing generalization and utilizing a large amount of unlabeled data. In fact, the three processing steps of our approach could be extended to incorporate self-supervised pretraining of the involved models. Despite these considerations, our method achieved the highest performance in the state of the art on the Teeth3DS benchmark.

Potential clinical application: The accuracy of this module is particularly crucial for applications such as crown and aligner design, where precise segmentation and labeling are essential to ensure a perfect fit and optimal function. Any inaccuracies at this stage can lead to suboptimal results, requiring additional adjustments and potentially increasing costs and time. In contrast, tasks like aligning 3D scans with other dental modalities (e.g., extraoral facial scans, CBCT images) are more tolerant of minor inaccuracies. Ultimately, the effectiveness of AI pipelines depends on the specific clinical application and the precision required by the algorithms used. Our proposed approach demonstrated the highest accuracy when compared to state-of-the-art methods in terms of teeth detection, segmentation, and labeling, making it a strong candidate for clinical use. Our model is well-suited for all AI workflows requiring teeth segmentation and labeling, such as crown and aligner

design, and multimodal dental data alignment. It can integrate with CAD software through various deployment options. Specifically, it can be implemented as a micro-web service, allowing for easy integration into existing systems via web-based interfaces. Alternatively, it can be deployed as an API, facilitating direct integration into standalone applications. This versatility ensures that our model can be seamlessly incorporated into diverse AI workflows and environments.

5. Conclusion

This paper proposes a novel deep learning-based approach for 3D teeth segmentation and labeling. The entire process is a three-stage pipeline starting with the coarse detection of the teeth. After that, the crown of each detected tooth is separated from the gum and the surrounding teeth. Finally, labels are assigned to the previously detected and segmented teeth crowns via a graph neural network model, which takes into account both teeth shape and teeth spatial arrangement clues. We qualitatively and quantitatively evaluated and compared our method with the state-of-the-art based on the Teeth3DS benchmark. Our approach yields superior results by significantly outperforming others in terms of teeth detection, segmentation, and labeling.

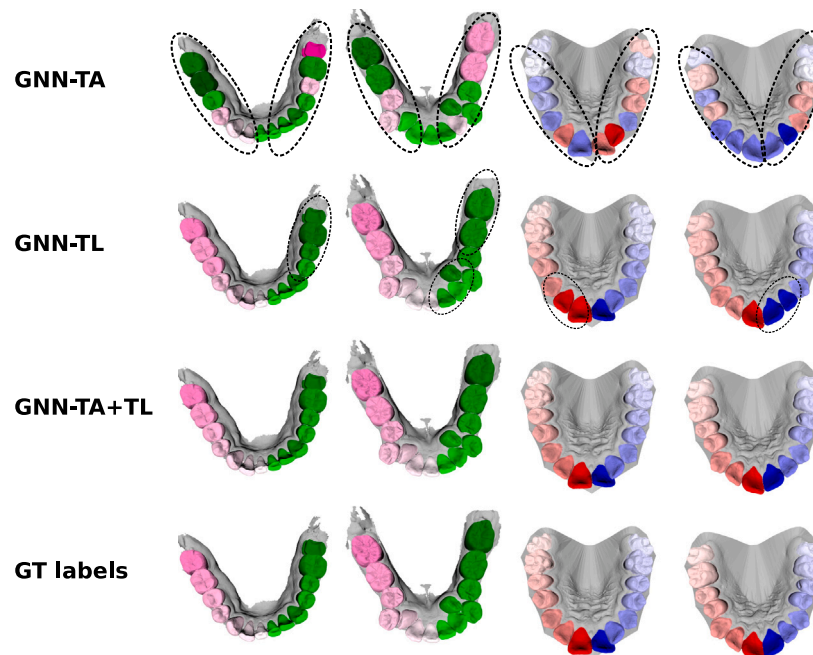


Fig. 16. Visual comparison of labeling results for four intraoral scans samples (columns) using different combinations of sub-networks (GNN-TA and GNN-TL) of our labeling model (rows).

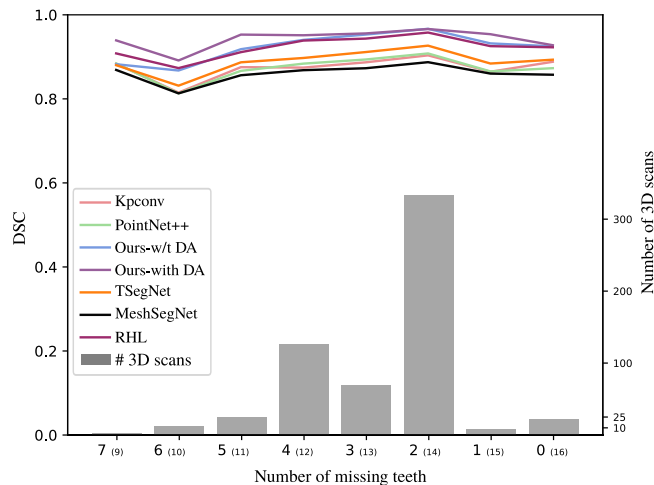


Fig. 17. Obtained DSC values for different missing teeth configurations in the testing dataset. The bars represent the number of 3D scans in the dataset corresponding to each missing teeth configuration. In addition to the number of missing teeth, we indicate the total number of teeth present in the scan for clarity. For example, 0 (16) indicates no missing teeth, representing a full arch with 16 teeth.

CRedit authorship contribution statement

Ahmed Rekik: Writing – review & editing, Software, Methodology, Investigation. **Achraf Ben-Hamadou:** Writing – review & editing, Supervision, Investigation. **Oussama Smaoui:** Writing – original draft, Software. **Firas Bouzguenda:** Software. **Sergi Pujades:** Supervision. **Edmond Boyer:** Supervision.

Declaration of competing interest

No potential conflicts of interest were disclosed by the authors.

References

- [1] Y. Kumar, R. Janardan, B. Larson, J. Moon, Improved segmentation of teeth in dental models, *Comput.-Aided Des. Appl.* 8 (2) (2011) 211–224.
- [2] M. Yaqi, L. Zhongke, Computer aided orthodontics treatment by virtual segmentation and adjustment, in: *IEEE International Conference on Image Analysis and Signal Processing, ICISP, 2010*, pp. 336–339.
- [3] B.-j. Zou, S.-j. Liu, S.-h. Liao, X. Ding, Y. Liang, Interactive tooth partition of dental mesh base on tooth-target harmonic field, *Comput. Biol. Med.* 56 (2015) 132–144.
- [4] J. Zhang, C. Li, Q. Song, L. Gao, Y.-K. Lai, Automatic 3D tooth segmentation using convolutional neural networks in harmonic parameter space, *Elsevier Graph. Model.* 39 (2020) 101071.
- [5] C. Lian, L. Wang, T. Wu, F. Wang, P. Yap, C. Ko, D. Shen, Deep multi-scale mesh feature learning for automated labeling of raw dental surfaces from 3D intraoral scanners, *IEEE Trans. Med. Imaging* 39 (7) (2020) 2440–2450.
- [6] Z. Cui, C. Li, N. Chen, G. Wei, R. Chen, Y. Zhou, W. Wang, TSegNet: An efficient and accurate tooth segmentation network on 3D dental model, *Med. Image Anal.* 69 (2020) 101949.
- [7] L. Qiu, C. Ye, P. Chen, Y. Liu, X. Han, S. Cui, DArch: Dental arch prior-assisted 3D tooth instance segmentation with weak annotations, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022*, pp. 20752–20761.
- [8] A. Ben-Hamadou, O. Smaoui, H. Chaabouni-Chouayakh, A. Rekik, S. Pujades, E. Boyer, J. Strippoli, A. Thollot, H. Setbon, C. Trosset, E. Ladroit, Teeth3DS: A benchmark for teeth segmentation and labeling from intra-oral 3D scans, 2022, arXiv:2210.06094.
- [9] T. Ciceri, L. Casartelli, F. Montano, S. Conte, L. Squarcina, A. Bertoldo, N. Agarwal, P. Brambilla, D. Peruzzo, Fetal brain mri atlases and datasets: A review, *NeuroImage* (2024) 120603.
- [10] J. Wang, F. Chen, Y. Ma, L. Wang, Z. Fei, J. Shuai, X. Tang, Q. Zhou, J. Qin, Xbound-former: Toward cross-scale boundary modeling in transformers, *IEEE Trans. Med. Imaging* 42 (6) (2023) 1735–1745.
- [11] H. Wu, J. Pan, Z. Li, Z. Wen, J. Qin, Automated skin lesion segmentation via an adaptive dual attention module, *IEEE Trans. Med. Imaging* 40 (1) (2020) 357–370.
- [12] S. Ding, J. Zheng, Z. Liu, Y. Zheng, Y. Chen, X. Xu, J. Lu, J. Xie, High-resolution dermoscopy image synthesis with conditional generative adversarial networks, *Biomed. Signal Process. Control* 64 (2021) 102224.
- [13] S. Pandey, P.R. Singh, J. Tian, An image augmentation approach using two-stage generative adversarial network for nuclei image segmentation, *Biomed. Signal Process. Control* 57 (2020) 101782.
- [14] C.-H. Wu, W.-H. Tsai, Y.-H. Chen, J.-K. Liu, Y.-N. Sun, Model-based orthodontic assessments for dental panoramic radiographs, *IEEE J. Biomed. Health Inform.* 22 (2) (2017) 545–551.

- [15] Y. Lai, F. Fan, Q. Wu, W. Ke, P. Liao, Z. Deng, H. Chen, Y. Zhang, Lcanet: Learnable connected attention network for human identification using dental images, *IEEE Trans. Med. Imaging* 40 (3) (2020) 905–915.
- [16] T.M. Tuan, H. Fujita, N. Dey, A.S. Ashour, V.T.N. Ngoc, D.-T. Chu, et al., Dental diagnosis from X-ray images: An expert system based on fuzzy computing, *Biomed. Signal Process. Control* 39 (2018) 64–73.
- [17] M. Rajee, C. Mythili, Gender classification on digital dental x-ray images using deep convolutional neural network, *Biomed. Signal Process. Control* 69 (2021) 102939.
- [18] W. Cui, Y. Wang, Y. Li, D. Song, X. Zuo, J. Wang, Y. Zhang, H. Zhou, B.s. Chong, L. Zeng, et al., Ctooth+: A large-scale dental cone beam computed tomography dataset and benchmark for tooth volume segmentation, in: *MICCAI Workshop on Data Augmentation, Labelling, and Imperfections*, Springer, 2022, pp. 64–73.
- [19] Y. Xiong, H. Zhang, S. Zhou, M. Lu, J. Huang, Q. Huang, B. Huang, J. Ding, Simultaneous detection of dental caries and fissure sealant in intraoral photos by deep learning: A pilot study, *BMC Oral. Health* 24 (1) (2024) 553.
- [20] S. Tian, M. Wang, F. Yuan, N. Dai, Y. Sun, W. Xie, J. Qin, Efficient computer-aided design of dental inlay restoration: a deep adversarial framework, *IEEE Trans. Med. Imaging* 40 (9) (2021) 2415–2427.
- [21] S. Tian, M. Wang, H. Ma, P. Huang, N. Dai, Y. Sun, J. Meng, Efficient tooth gingival margin line reconstruction via adversarial learning, *Biomed. Signal Process. Control* 78 (2022) 103954.
- [22] S. Shujaat, M.M. Bornstein, J.B. Price, R. Jacobs, Integration of imaging modalities in digital dental workflows-possibilities, limitations, and potential future developments, *Dentomaxillofacial Radiol.* 50 (7) (2021) 20210268.
- [23] X. Chen, N. Ma, T. Xu, C. Xu, Deep learning-based tooth segmentation methods in medical imaging: A review, *Proc. Inst. Mech. Eng. H* 238 (2) (2024) 115–131.
- [24] M. Zhao, L. Ma, W. Tan, D. Nie, Interactive tooth segmentation of dental models, in: *IEEE Engineering in Medicine and Biology Conference, IMBC'06*, IEEE, 2006, pp. 654–657.
- [25] T. Yuan, W. Liao, N. Dai, X. Cheng, Q. Yu, Single-tooth modeling for 3D dental model, *Int. J. Biomed. Imag.* 2010 (2010).
- [26] T. Kronfeld, D. Brunner, G. Brunnett, Snake-based segmentation of teeth from virtual dental casts, *Comput.-Aided Des. Appl.* 7 (2) (2010) 221–233.
- [27] C. Sinthanayothin, W. Tharanont, Orthodontics treatment simulation by teeth segmentation and setup, in: *International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology, ECTICON'08*, vol. 1, IEEE, 2008, pp. 81–84.
- [28] S.-h. Liao, S.-j. Liu, B.-j. Zou, X. Ding, Y. Liang, J.-h. Huang, Automatic tooth segmentation of dental mesh based on harmonic fields, *BioMed. Res. Int.* 2015 (2015).
- [29] Z. Zou, K. Chen, Z. Shi, Y. Guo, J. Ye, Object detection in 20 years: A survey, *Proc. IEEE* (2023).
- [30] Z. Cui, C. Li, W. Wang, ToothNet: Automatic tooth instance segmentation and identification from cone beam CT images, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 6368–6377.
- [31] K. He, G. Gkioxari, P. Dollár, R. Girshick, Mask R-CNN, in: *2017 IEEE International Conference on Computer Vision, ICCV*, 2017, pp. 2980–2988.
- [32] Y. Rao, Y. Wang, F. Meng, J. Pu, J. Sun, Q. Wang, A symmetric fully convolutional residual network with DCRF for accurate tooth segmentation, *IEEE Access* 8 (2020) 92028–92038.
- [33] M. Do Nascimento Gerhardt, R.C. Fontenele, A.F. Leite, P. Lahoud, A. Van Gerven, H. Willems, A. Smolders, T. Beznik, R. Jacobs, Automated detection and labelling of teeth and small edentulous regions on cone-beam computed tomography using convolutional neural networks, *J. Dent.* 122 (2022) 104139.
- [34] D. Sun, Y. Pei, G. Song, Y. Guo, G. Ma, T. Xu, H. Zha, Tooth segmentation and labeling from digital dental casts, in: *IEEE International Symposium on Biomedical Imaging, ISBI'20*, 2020.
- [35] N. Verma, E. Boyer, J. Verbeek, FeaStNet: Feature-steered graph convolutions for 3D shape analysis, in: *IEEE Conference on Computer Vision and Pattern Recognition, CVPR*, 2018, pp. 2598–2606.
- [36] D. Sun, Y. Pei, P. Li, G. Song, Y. Guo, H. Zha, T. Xu, Automatic tooth segmentation and dense correspondence of 3D dental model, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2020, pp. 703–712.
- [37] X. Xu, C. Liu, Y. Zheng, 3D tooth segmentation and labeling using deep convolutional neural networks, *IEEE Trans. Vis. Comput. Graphics* 25 (7) (2018) 2336–2348.
- [38] F.G. Zanjani, D.A. Moin, B. Verheij, F. Claessen, T. Cherici, T. Tan, et al., Deep learning approach to semantic segmentation in 3D point cloud intra-oral scans of teeth, in: *International Conference on Medical Imaging with Deep Learning, MIDL*, 2019.
- [39] C.R. Qi, H. Su, K. Mo, L.J. Guibas, PointNet: Deep learning on point sets for 3D classification and segmentation, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 652–660.
- [40] S. Tian, N. Dai, B. Zhang, F. Yuan, Q. Yu, X. Cheng, Automatic classification and segmentation of teeth on 3D dental model using hierarchical deep learning networks, *IEEE Access* 7 (2019) 84817–84828.
- [41] F.G. Zanjani, D.A. Moin, F. Claessen, T. Cherici, S. Parinussa, A. Pourtaherian, S. Zinger, et al., Mask-mcnet: Instance segmentation in 3D point cloud of intra-oral scans, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention, MICCAI'19*, Springer, 2019, pp. 128–136.
- [42] Q. Ma, G. Wei, Y. Zhou, X. Pan, S. Xin, W. Wang, SRF-Net: Spatial relationship feature network for tooth point cloud classification, *Comput. Graph. Forum* 39 (7) (2020) 267–277.
- [43] Y. Zhao, L. Zhang, C. Yang, Y. Tan, Y. Liu, P. Li, T. Huang, C. Gao, 3D dental model segmentation with graph attentional convolution network, *Pattern Recognit. Lett.* 152 (2021) 79–85.
- [44] Y. Zhao, L. Zhang, Y. Liu, D. Meng, Z. Cui, C. Gao, X. Gao, C. Lian, D. Shen, Two-stream graph convolutional network for intra-oral scanner image segmentation, *IEEE Trans. Med. Imag.* 41 (2022-04).
- [45] Y. Wang, Y. Sun, Z. Liu, S.E. Sarma, M.M. Bronstein, J.M. Solomon, Dynamic graph CNN for learning on point clouds, *ACM Trans. Graph.* (2019).
- [46] L. Krenmayr, R. von Schwerin, D. Schaudt, P. Riedel, A. Hafner, DilatedToothSegNet: Tooth segmentation network on 3D dental meshes through increasing receptive vision, *J. Imag. Inform. Med.* (2024) 1–17.
- [47] S. Zhuang, G. Wei, Z. Cui, Y. Zhou, Robust hybrid learning for automatic teeth segmentation and labeling on 3D dental models, *IEEE Trans. Multimed.* (2023) 1–13.
- [48] A. Ben-Hamadou, O. Smaoui, A. Rekik, S. Pujades, E. Boyer, H. Lim, M. Kim, M. Lee, M. Chung, Y.-G. Shin, M. Leclercq, L. Cevdianes, J.C. Prieto, S. Zhuang, G. Wei, Z. Cui, Y. Zhou, T. Dascalu, B. Ibragimov, T.-H. Yong, H.-G. Ahn, W. Kim, J.-H. Han, B. Choi, N. van Nistelrooij, S. Kempers, S. Vinayahalingam, J. Strippoli, A. Thollot, H. Setbon, C. Trosset, E. Ladroit, 3DTeethSeg'22: 3D teeth scan segmentation and labeling challenge, 2023, arXiv:2305.18277.
- [49] M.-H. Guo, J.-X. Cai, Z.-N. Liu, T.-J. Mu, R.R. Martin, S.-M. Hu, Pct: Point cloud transformer, *Comput. Vis. Media* 7 (2) (2021) 187–199.
- [50] J. Jeong, Y. Lei, S. Kahn, T. Liu, W.J. Curran, H.-K. Shu, H. Mao, X. Yang, Brain tumor segmentation using 3D mask R-CNN for dynamic susceptibility contrast enhanced perfusion imaging, *Phys. Med. Biol.* 65 (18) (2020) 185009.
- [51] M. Eck, T. DeRose, T. Duchamp, H. Hoppe, M. Lounsbery, W. Stuetzle, Multiresolution analysis of arbitrary meshes, in: *Proceedings of the 22nd Annual Conference on Computer Graphics and Interactive Techniques*, in: *SIGGRAPH 95*, Association for Computing Machinery, New York, NY, USA, 1995, pp. 173–182.
- [52] M. Fey, J.E. Lenssen, F. Weichert, H. Müller, SplineCNN: Fast geometric deep learning with continuous B-spline kernels, in: *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, IEEE, 2018, pp. 869–877.
- [53] P. Chlap, H. Min, N. Vandenberg, J. Dowling, L. Holloway, A. Haworth, A review of medical image data augmentation techniques for deep learning applications, *J. Med. Imag. Radiat. Oncol.* 65 (5) (2021) 545–563.
- [54] T.H. Farook, S. Ahmed, J. Giri, F. Rashid, T. Hughes, J. Dudley, Influence of intraoral scanners, operators, and data processing on dimensional accuracy of dental casts for unsupervised clinical machine learning: An in vitro comparative study, *Int. J. Dent.* 2023 (1) (2023) 7542813.
- [55] H. Thomas, C.R. Qi, J.-E. Deschaud, B. Marcotequi, F. Goulette, L.J. Guibas, Kpconv: Flexible and deformable convolution for point clouds, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 6411–6420.
- [56] R.R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-CAM: Visual explanations from deep networks via gradient-based localization, *Int. J. Comput. Vis.* 128 (2020) 336–359.